

Vietnam Weather Forecast Dashboard

1. Project Overview

Vietnam Weather Forecast Dashboard is a machine learning based weather forecasting system for 34 provinces and cities in Vietnam.

The project combines:

- historical weather data collection
- automated dataset synchronization
- machine learning model training
- forecast evaluation
- FastAPI backend services
- React + TypeScript frontend visualization

The main objective is to predict whether rain will occur in future days using historical weather observations.

The project supports:

- 1-day forecast
- 3-day forecast
- 7-day forecast
- historical evaluation mode
- exploratory weather analysis
- automatic weather dataset updates

2. System Architecture

The project is divided into 3 main layers.

```
graph TD; A[Open Meteo API] --> B[Data Fetching + Preprocessing]; B --> C[Machine Learning Training Pipeline]; C --> D[FastAPI Backend]; D --> E[React Frontend Dashboard];
```

3. Weather Data Source

3.1 Main Weather Provider

The project uses weather data from Open Meteo.

Main website:

```
https://open-meteo.com/
```

Historical archive API:

```
https://archive-api.open-meteo.com/v1/archive
```

The system fetches historical daily weather observations for all supported Vietnamese provinces.

3.2 Weather Variables Used

The project collects the following weather features:

Feature	Description	Unit
temperature_2m_mean	Mean daily temperature	°C
relative_humidity_2m_mean	Mean daily humidity	%
wind_speed_10m_max	Maximum daily wind speed	km/h
cloud_cover_mean	Mean daily cloud cover	%
surface_pressure_mean	Mean surface pressure	hPa
precipitation_sum	Total precipitation	mm

3.3 Rain Label Generation

The API returns:

```
precipitation_sum
```

The project converts precipitation into binary rain labels.

Condition	Rain Label
precipitation_sum > 0	1
precipitation_sum = 0	0

This transforms the forecasting problem into binary classification.

3.4 Historical Data Fetch URL

The backend dynamically generates weather API requests.

Example:

```
url = (
    "https://archive-api.open-meteo.com/v1/archive?"
    f"latitude={latitude}"
    f"&longitude={longitude}"
    f"&start_date={start_date}"
    f"&end_date={end_date}"
    "&daily="
    "temperature_2m_mean,"
    "relative_humidity_2m_mean,"
    "wind_speed_10m_max,"
    "cloud_cover_mean,"
    "surface_pressure_mean,"
    "precipitation_sum"
    "&timezone=Asia%2FBangkok"
)
```

3.5 Example API Request

Example request for Ho Chi Minh City:

Field	Value
Province	Ho Chi Minh
Latitude	10.8231
Longitude	106.6297

Example API URL:

```
https://archive-api.open-meteo.com/v1/archive?
latitude=10.8231&longitude=106.6297&start_date=2025-01-01&end_date=2025-01-31&daily=temperatur
```

3.6 Example API Response

The API returns JSON formatted weather data.

Example:

```
{
  "latitude": 10.790861,
  "longitude": 106.6313,
  "generationtime_ms": 5.9117,
  "timezone": "GMT",
  "daily_units": {
    "time": "iso8601",
    "temperature_2m_mean": "°C",
    "precipitation_sum": "mm"
  },
  "daily": {
    "time": [
      "2025-01-01",
      "2025-01-02",
      "2025-01-03"
    ],
    "temperature_2m_mean": [
      25.5,
      26.3,
      26.2
    ],
    "precipitation_sum": [
      0.00,
      0.30,
      0.80
    ]
  }
}
```

The backend then converts this response into structured pandas DataFrames.

3.7 Internal Dataset Format

After preprocessing, each province dataset is stored as:

date	destination	latitude	longitude	temp	humidity	wind_speed	cloud_cover	pressure	ra
2020-01-01	Ha_Noi	21.0285	105.8542	20.0	88	9.2	83	1021.4	1

The backend stores one dataset file per province.

4. Supported Provinces

The project currently supports 34 provinces and cities in Vietnam.

Province	Latitude	Longitude
Ha_Noi	21.0285	105.8542
Hue	16.4637	107.5909
Lai_Chau	22.3964	103.4582
Dien_Bien	21.3860	103.0230
Son_La	21.3256	103.9188
Lang_Son	21.8537	106.7610
Quang_Ninh	21.0064	107.2925
Thanh_Hoa	19.8067	105.7852
Nghe_An	18.6796	105.6813
Ha_Tinh	18.3550	105.8877
Cao_Bang	22.6667	106.2500
Tuyen_Quang	21.8233	105.2181
Lao_Cai	22.4856	103.9707
Thai_Nguyen	21.5942	105.8482
Phu_Tho	21.3227	105.4010
Bac_Ninh	21.1861	106.0763
Hung_Yen	20.6464	106.0511
Hai_Phong	20.8449	106.6881
Ninh_Binh	20.2506	105.9745
Quang_Tri	16.7500	107.2000
Da_Nang	16.0544	108.2022
Quang_Ngai	15.1214	108.8044
Gia_Lai	13.9833	108.0000

Province	Latitude	Longitude
Khanh_Hoa	12.2388	109.1967
Lam_Dong	11.9404	108.4583
Dak_Lak	12.7100	108.2378
Ho_Chi_Minh	10.8231	106.6297
Dong_Nai	10.9453	106.8240
Tay_Ninh	11.3352	106.1099
Can_Tho	10.0452	105.7469
Vinh_Long	10.2537	105.9722
Dong_Thap	10.4938	105.6881
Ca_Mau	9.1769	105.1524
An_Giang	10.5216	105.1259

5. Dataset Timeline

Training Dataset

Start Date	End Date
2020-01-01	2025-12-31

Testing Dataset

Start Date	End Date
2026-01-01	2026-04-30

The testing dataset is separated from training data to avoid leakage.

6. Feature Engineering

6.1 Lookback Window

The project uses rolling historical windows.

Example configuration:

LOOKBACKS = [3, 5, 7, 10, 15, 30]

A lookback window means:

- use previous historical days
- convert them into lag features
- predict future rain occurrence

Example:

For lookback = 7:

The model receives:

- temperature from previous 7 days
- humidity from previous 7 days
- pressure from previous 7 days
- cloud cover from previous 7 days
- wind speed from previous 7 days

and predicts future rain labels.

6.2 Forecast Horizons

The system supports:

Horizon	Meaning
1	predict current / next day
3	predict next 3 days
7	predict next 7 days

Each horizon has independent trained models.

7. Machine Learning Pipeline

7.1 Problem Type

The project is a binary classification problem.

Label	Meaning
0	no rain

Label	Meaning
1	rain

7.2 Models Used

The project trains and evaluates:

Model	Description
Logistic Regression	linear classifier
Random Forest	ensemble tree model
XGBoost	gradient boosting
LightGBM	gradient boosting

7.3 Training Strategy

Models are trained separately for:

- province
- forecast horizon
- lookback range
- algorithm type

This creates:

province × horizon × lookback × algorithm

trained model combinations.

7.4 Example Training Flow

```
Load Province Dataset
↓
Generate Lag Features
↓
Split Train/Test Data
↓
Train ML Model
↓
```


Generate Predictions



Evaluate Metrics



Save Best Model

8. Model Evaluation

The project evaluates:

Metric	Description
Accuracy	overall correctness
Precision	rain prediction quality
Recall	rain detection sensitivity
F1 Score	balance of precision and recall
ROC AUC	classification quality

The backend automatically ranks models using:

- F1 score
- ROC AUC
- Accuracy

and selects the best model.

9. Backend Architecture

The backend is implemented using FastAPI.

Main responsibilities:

- weather data synchronization
 - prediction API
 - evaluation API
 - model loading
 - feature generation
 - EDA snapshot generation
-

9.1 Main API Endpoints

Endpoint	Purpose
GET /api/health	health check
GET /api/provinces	list supported provinces
POST /api/fetch-all	synchronize datasets
POST /api/exploration	exploratory analysis
POST /api/predict	prediction endpoint

10. Frontend Architecture

The frontend is built using:

- React
- TypeScript
- Vite
- TailwindCSS
- Recharts

10.1 Frontend Features

Forecast Mode

Users can:

- select province
- select algorithm
- select forecast horizon
- generate future predictions

Evaluation Mode

Users can:

- choose historical dates
 - compare prediction vs actual rain labels
 - compute prediction correctness
-

EDA Dashboard

The frontend visualizes:

- temperature distributions
- humidity distributions
- cloud cover distributions
- wind speed distributions
- pressure distributions
- rain distributions

for all provinces.

11. Automatic Dataset Synchronization

The backend automatically updates datasets.

Example:

If the newest local dataset date is:

2025-12-31

and today's date is:

2026-05-20

then the system fetches:

2026-01-01 → 2026-05-19

for all provinces.

The current day is excluded because weather observations may still be incomplete.

12. Forecast Workflow

```
graph TD
    A[Frontend Request] --> B[FastAPI Backend]
    B --> C[Load Province Dataset]
```

```
↓
Load Best Model
↓
Generate Features
↓
Predict Rain Probability
↓
Return Prediction Response
```

13. Evaluation Workflow

```
Historical Date Selected
↓
Generate Prediction
↓
Load Actual Weather Result
↓
Compare Prediction vs Actual
↓
Compute Correctness
```

14. Project Structure

```
backend/
├─ fetch_data/
├─ load_weather_data/
├─ ml_pipeline/
├─ models/
├─ utils/
└─ main.py

frontend/
├─ src/
├─ app/
├─ components/
└─ styles/
```

15. Technologies Used

Backend

- Python
 - FastAPI
 - pandas
 - scikit-learn
 - XGBoost
 - LightGBM
 - joblib
-

Frontend

- React
 - TypeScript
 - Vite
 - TailwindCSS
 - Recharts
-

16. Current Limitations

Current limitations include:

- only 34 provinces supported
 - daily weather data only
 - no satellite imagery
 - no hourly forecasting
 - no deep learning models
 - no GPU acceleration
 - no cloud deployment
-

17. Future Improvements

Potential future improvements:

- LSTM forecasting
- Transformer forecasting
- ensemble learning
- anomaly detection
- automatic retraining pipeline
- Docker deployment
- CI/CD integration
- PostgreSQL storage

- cloud hosting
 - real-time streaming weather updates
-

18. Conclusion

This project demonstrates a complete end-to-end weather forecasting platform for Vietnam.

The system integrates:

- weather data collection
- preprocessing
- feature engineering
- machine learning training
- model evaluation
- REST API services
- frontend visualization

into a unified forecasting dashboard.